

SEMESTER PROJECT REPORT

*Design and Configuration of a Multi-Campus Network
for The University of Faisalabad*

Program	BSIT — 6th Semester
Subject	Computer Networks
Tool Used	Cisco Packet Tracer
Campuses	Amin Campus Saleem Campus
Submitted To	Sir Ibrar
Date	2026

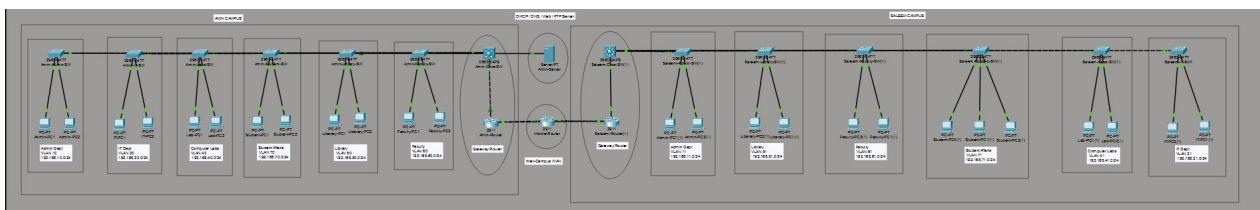
1. Introduction

This report documents the complete design and implementation of a campus network for The University of Faisalabad (TUF) using Cisco Packet Tracer. The network spans two campuses — Amin Campus and Saleem Campus — connected through a dedicated WAN link.

The design follows a hierarchical model: each campus has one gateway router, one Layer 3 core switch, and multiple access-layer switches — one per department. VLANs are used for departmental segmentation, 802.1Q trunking carries tagged traffic between switches and routers, and OSPF provides dynamic routing between campuses. A centralized DHCP server on Amin Campus serves all VLANs on both campuses via ip helper-address.

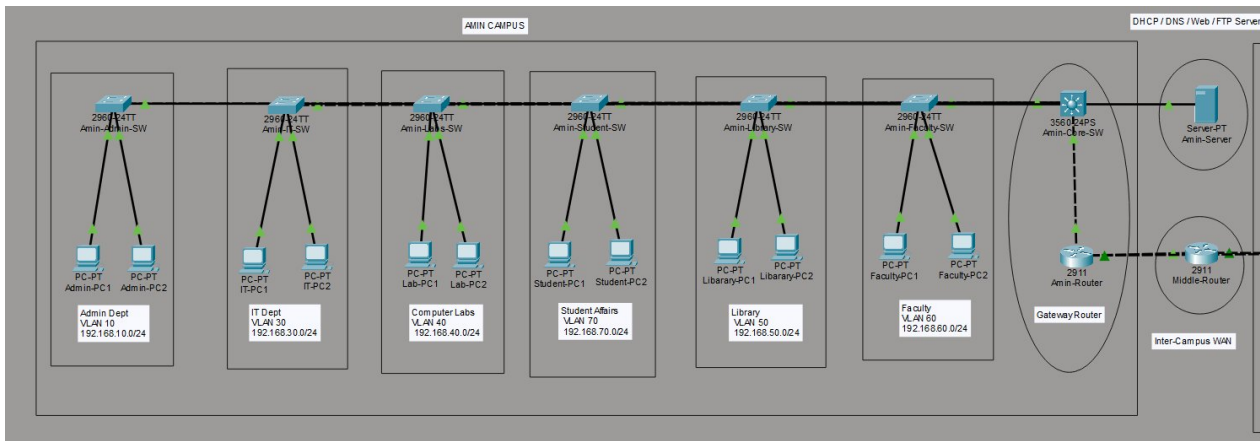
2. Network Topology

The topology below shows both campuses connected via a WAN cloud. Amin Campus sits on the left; Saleem Campus on the right. The centralized DHCP/DNS/Web/FTP server is hosted on Amin Campus.



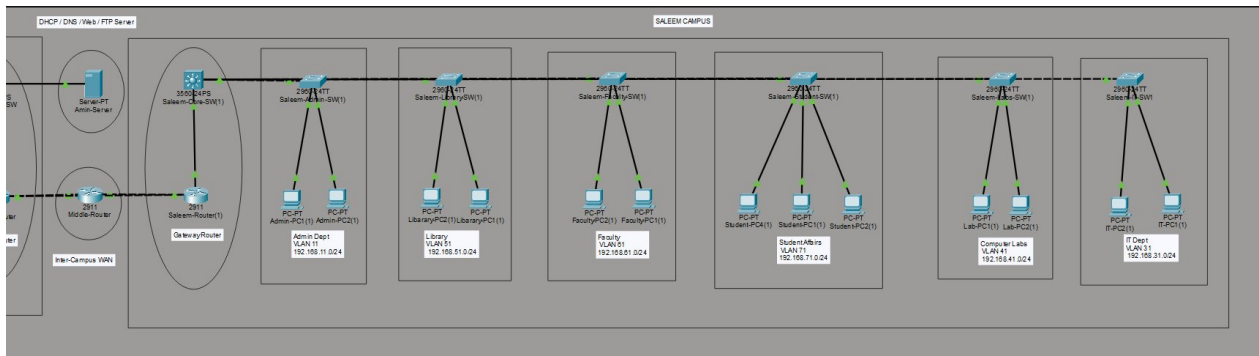
2.1 Amin Campus

Amin Campus consists of Amin-Router (Cisco 2911), Amin-Core-SW (Cisco 3560), six Cisco 2960 access switches (Admin, IT, Labs, Library, Faculty, Students), and Amin-Server hosting DHCP, DNS, Web, and FTP services.



2.2 Saleem Campus

Saleem Campus mirrors the same structure: Saleem-Router (Cisco 2911), Saleem-Core-SW (Cisco 3560), and six Cisco 2960 access switches for the same six departments.



3. VLAN Design

VLANs logically separate departmental traffic across both campuses. Amin Campus uses even-decade IDs (10, 20, 30 ...); Saleem Campus uses the next consecutive IDs (11, 31, 41 ...).

VLAN ID (Amin / Saleem)	Department	Subnet (Amin)	Subnet (Saleem)
10 / 11	Administration	192.168.10.0/24	192.168.11.0/24
20 / —	Accounts	192.168.20.0/24	—
30 / 31	IT Department	192.168.30.0/24	192.168.31.0/24
40 / 41	Computer Labs	192.168.40.0/24	192.168.41.0/24
50 / 51	Library	192.168.50.0/24	192.168.51.0/24
60 / 61	Faculty	192.168.60.0/24	192.168.61.0/24
70 / 71	Students/Wireless	192.168.70.0/24	192.168.71.0/24
80 / —	Servers (Amin)	192.168.80.0/24	—

4. IP Addressing Scheme

4.1 Amin Campus Hosts

Device	IP Address	Subnet Mask	Default Gateway
Amin-Server	192.168.80.10	255.255.255.0	192.168.80.1
Admin-PC1	192.168.10.x	255.255.255.0	192.168.10.1
Admin-PC2	192.168.10.x	255.255.255.0	192.168.10.1
IT-PC1	192.168.30.x	255.255.255.0	192.168.30.1
IT-PC2	192.168.30.x	255.255.255.0	192.168.30.1
Lab-PC1	192.168.40.x	255.255.255.0	192.168.40.1
Library-PC1	192.168.50.x	255.255.255.0	192.168.50.1
Faculty-PC1	192.168.60.x	255.255.255.0	192.168.60.1
Student-PC1	192.168.70.x	255.255.255.0	192.168.70.1

4.2 Saleem Campus Hosts

Device	IP Address	Subnet Mask	Default Gateway
Admin-PC1(1)	192.168.11.x	255.255.255.0	192.168.11.1
Admin-PC2(1)	192.168.11.x	255.255.255.0	192.168.11.1

IT-PC1(1)	192.168.31.x	255.255.255.0	192.168.31.1
Lab-PC1(1)	192.168.41.x	255.255.255.0	192.168.41.1
Library-PC1(1)	192.168.51.x	255.255.255.0	192.168.51.1
Faculty-PC1(1)	192.168.61.12	255.255.255.0	192.168.61.1
Faculty-PC2(1)	192.168.61.x	255.255.255.0	192.168.61.1
Student-PC1(1)	192.168.71.x	255.255.255.0	192.168.71.1

4.3 WAN Link

Interface	IP Address	Description
Amin-Router Gi0/1	10.10.10.1/30	WAN — Amin Side
Middle-Router Gi0/0	10.10.10.2/30	WAN — Middle
Middle-Router Gi0/1	10.10.10.9/30	WAN — Saleem Side
Saleem-Router Gi0/2	10.10.10.10/30	WAN — Saleem Side

5. Router Configuration

5.1 Inter-VLAN Routing — Sub-Interfaces

Both routers use router-on-a-stick (802.1Q sub-interfaces on Gi0/0) to route between VLANs. Each sub-interface is assigned the gateway IP for that VLAN. ip helper-address 192.168.80.10 is configured on each Amin-Router sub-interface to relay DHCP requests to Amin-Server.

Sub-Interface	VLAN	IP (Gateway)	ip helper-address
Gi0/0.10	10	192.168.10.1	192.168.80.10
Gi0/0.20	20	192.168.20.1	192.168.80.10
Gi0/0.30	30	192.168.30.1	192.168.80.10
Gi0/0.40	40	192.168.40.1	192.168.80.10
Gi0/0.50	50	192.168.50.1	192.168.80.10
Gi0/0.60	60	192.168.60.1	192.168.80.10
Gi0/0.70	70	192.168.70.1	192.168.80.10
Gi0/0.80	80	192.168.80.1	—

5.2 Saleem-Router CLI — Sub-Interfaces Active



```
Saleem-Router(1)
Physical Config CLI Attributes

This product contains cryptographic features and is subject to United States and local country law governing import, export, transfer and use. Delivery of Cisco cryptographic products does not imply third-party authority to import, export, distribute or use encryption. Importers, exporters, distributors and users are responsible for compliance with U.S. and local country law. By using this product you agree to comply with applicable laws and regulations. If you are unable to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wf/public/exports/crypto/faq.html
If you require further assistance please contact us by sending email to:
export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FX152400K3
0 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
16384 bytes of non-volatile configuration memory.
149886K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/10, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/20, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/30, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/40, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/50, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/60, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/70, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/80, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
VLINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
16:02:32: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on GigabitEthernet0/2 from LOADING to FULL, Loading Done
16:02:37: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

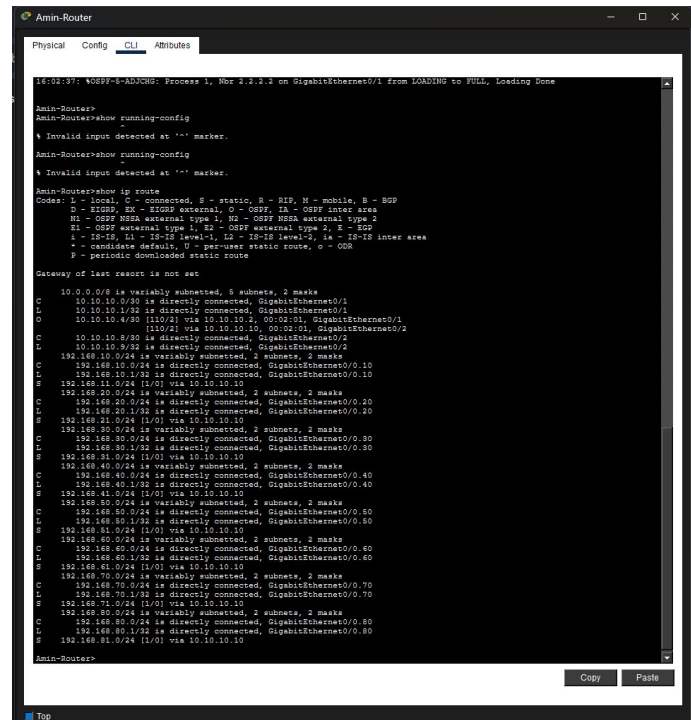
Saleem-Router>show ip ospf neighbor

Neighbor ID Pri State Dead Time Address Interface
2.2.2.2 1 FULL/DR 00:00:32 10.10.10.6 GigabitEthernet0/1
1.1.1.1 1 FULL/DR 00:00:32 10.10.10.9 GigabitEthernet0/2
Saleem-Router>
Saleem-Router>show running-config
% Invalid input detected at '^' marker.
Saleem-Router#
```

5.3 OSPF Routing

OSPF (Area 0) is configured on Amin-Router, Middle-Router, and Saleem-Router. All VLAN subnets and WAN links are advertised. The show ip ospf neighbor output confirms FULL adjacency between all

three routers.



```
16:01:37: %OSPF-4-ADJRC: Process 1, Nbr 1.2.2.2 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

Amin-Router>
Amin-Router#show running-config
-
% Invalid input detected at '^' marker.
Amin-Router#show running-config
-
% Invalid input detected at '^' marker.

Amin-Router#show ip route
Codes: C - local, S - connected, S* - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, I - ISP
        I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, IS - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

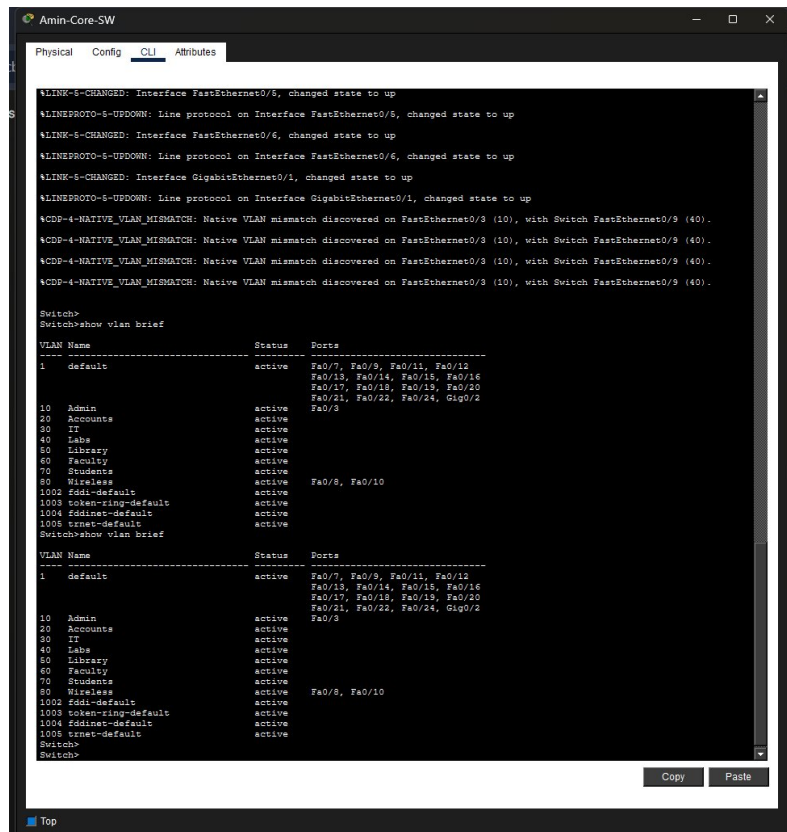
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C    10.10.10.0/30 is directly connected, GigabitEthernet0/1
L    10.10.10.1/32 is directly connected, GigabitEthernet0/1
O    10.10.10.4/30 [110/2] via 10.10.10.3, 00:02:01, GigabitEthernet0/1
C    10.10.10.8/30 [110/2] via 10.10.10.10, 00:02:01, GigabitEthernet0/2
L    10.10.10.9/32 is directly connected, GigabitEthernet0/2
C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
R    192.168.11.0/24 [1/0] via 10.10.10.10
C    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L    192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
R    192.168.41.0/24 [1/0] via 10.10.10.10
C    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L    192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
R    192.168.40.0/24 [1/0] via 10.10.10.10
C    192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L    192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
R    192.168.41.0/24 [1/0] via 10.10.10.10
C    192.168.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.50.0/24 is directly connected, GigabitEthernet0/0.50
L    192.168.50.1/32 is directly connected, GigabitEthernet0/0.50
R    192.168.60.0/24 [1/0] via 10.10.10.10
C    192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.60.0/24 is directly connected, GigabitEthernet0/0.60
L    192.168.60.1/32 is directly connected, GigabitEthernet0/0.60
R    192.168.61.0/24 [1/0] via 10.10.10.10
C    192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.70.0/24 is directly connected, GigabitEthernet0/0.70
L    192.168.70.1/32 is directly connected, GigabitEthernet0/0.70
R    192.168.80.0/24 [1/0] via 10.10.10.10
C    192.168.80.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.80.0/24 is directly connected, GigabitEthernet0/0.80
L    192.168.80.1/32 is directly connected, GigabitEthernet0/0.80
R    192.168.81.0/24 [1/0] via 10.10.10.10

Amin-Router>
```

5.4 Amin-Router Routing Table

The routing table on Amin-Router shows all directly connected VLANs (C) and all OSPF-learned Saleem Campus subnets (S/O), confirming end-to-end reachability.



```
16:01:37: %LINK-6-CHANGED: Interface FastEthernet0/5, changed state to up
16:01:37: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
16:01:37: %LINK-6-CHANGED: Interface FastEthernet0/6, changed state to up
16:01:37: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
16:01:37: %LINK-6-CHANGED: Interface GigabitEthernet0/1, changed state to up
16:01:37: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with Switch FastEthernet0/3 (40).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with Switch FastEthernet0/3 (40).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with Switch FastEthernet0/3 (40).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with Switch FastEthernet0/3 (40).

Switch>
Switch>show vlan brief

VLAN Name                Status    Ports
----
1    default                active    Fa0/7, Fa0/9, Fa0/11, Fa0/12,
                                Fa0/13, Fa0/14, Fa0/15, Fa0/16,
                                Fa0/17, Fa0/18, Fa0/19, Fa0/20,
                                Fa0/21, Fa0/22, Fa0/24, Gig0/2
10   Admin                  active
20   Accounts               active
30   IT                     active
40   Labs                   active
50   Library                active
60   Faculty                active
70   Students               active
80   Wireless               active
1002 fddi-default           active
1003 tokenring-default     active
1004 fddinet-default       active
1005 trnet-default         active

Switch>show vlan brief

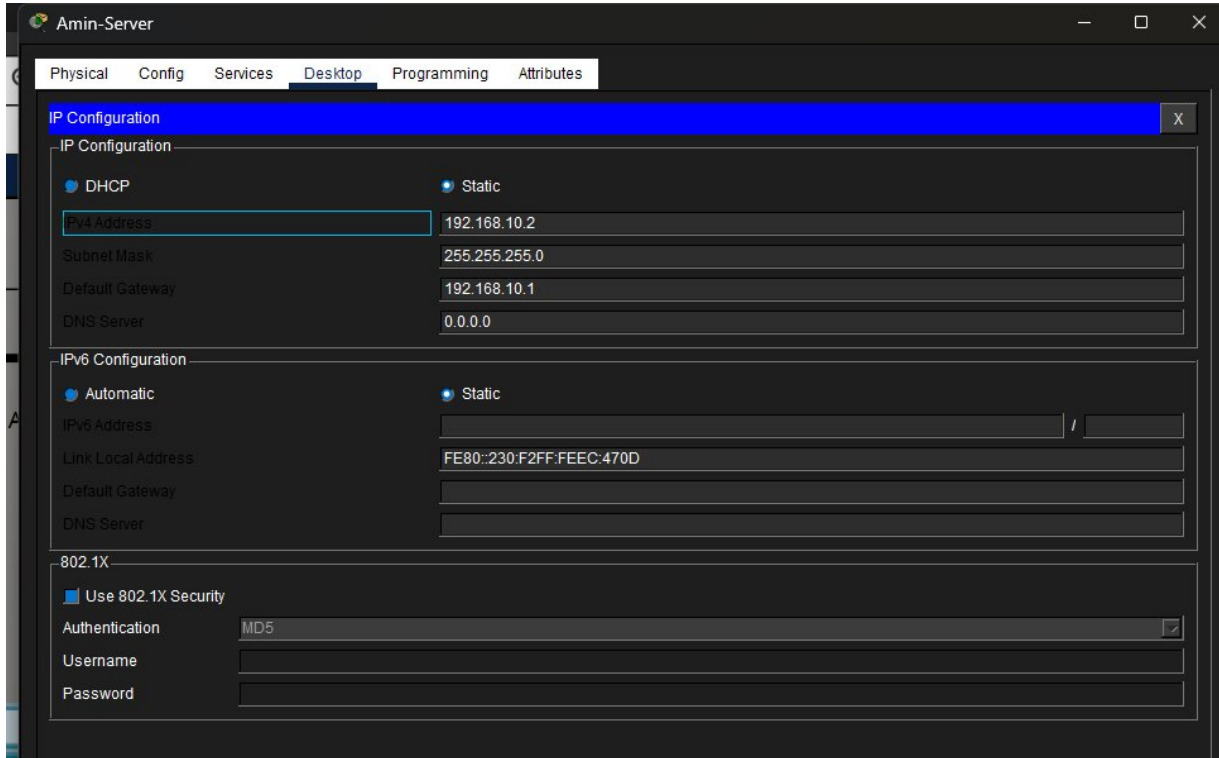
VLAN Name                Status    Ports
----
1    default                active    Fa0/7, Fa0/9, Fa0/11, Fa0/12,
                                Fa0/13, Fa0/14, Fa0/15, Fa0/16,
                                Fa0/17, Fa0/18, Fa0/19, Fa0/20,
                                Fa0/21, Fa0/22, Fa0/24, Gig0/2
10   Admin                  active
20   Accounts               active
30   IT                     active
40   Labs                   active
50   Library                active
60   Faculty                active
70   Students               active
80   Wireless               active
1002 fddi-default           active
1003 tokenring-default     active
1004 fddinet-default       active
1005 trnet-default         active

Switch>
```

6. Switch Configuration

6.1 VLAN Verification — Amin-Core-SW

The show vlan brief command on Amin-Core-SW confirms all VLANs (10, 20, 30, 40, 50, 60, 70, 80) are active and assigned to the correct ports.



6.2 Access Switch Port Assignment

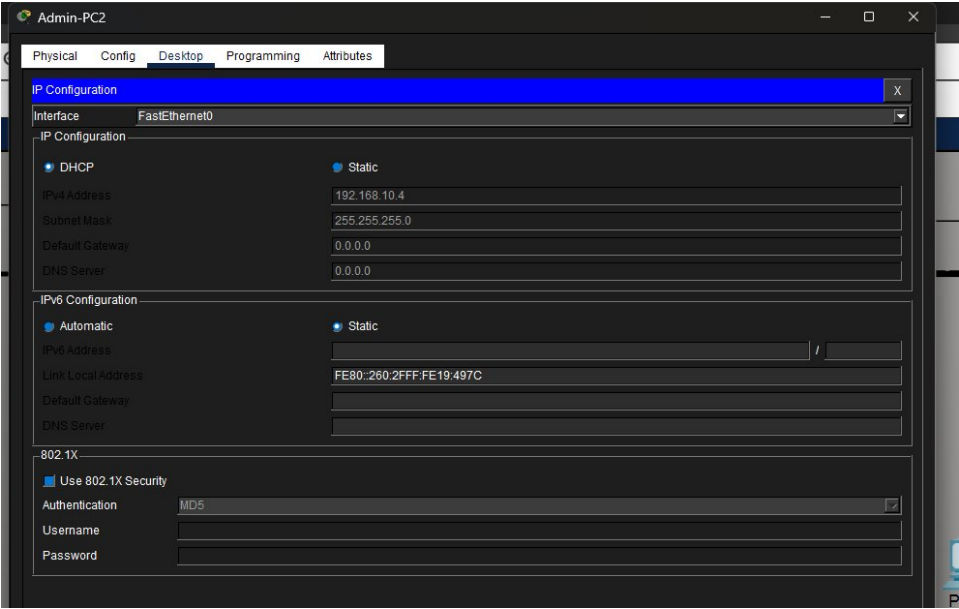
Switch	VLAN	Access Ports	Trunk Port
Amin-Admin-SW	10	Fa0/1–2	Fa0/9 → Core
Amin-IT-SW	30	Fa0/1–2	Fa0/9 → Core
Amin-Labs-SW	40	Fa0/1–2	Fa0/9 → Core
Amin-Library-SW	50	Fa0/1–2	Fa0/9 → Core
Amin-Faculty-SW	60	Fa0/1–2	Fa0/9 → Core
Amin-Student-SW	70	Fa0/1–2	Fa0/9 → Core
Saleem-Admin-SW(1)	11	Fa0/1–2	Fa0/5 → Core
Saleem-IT-SW(1)	31	Fa0/1–2	Fa0/5 → Core
Saleem-Labs-SW(1)	41	Fa0/1–2	Fa0/9 → Core
Saleem-Library-SW(1)	51	Fa0/1–2	Fa0/9 → Core
Saleem-Faculty-SW(1)	61	Fa0/1–2	Fa0/9 → Core

Saleem-Student-SW(1)	71	Fa0/1–2	Fa0/9 → Core
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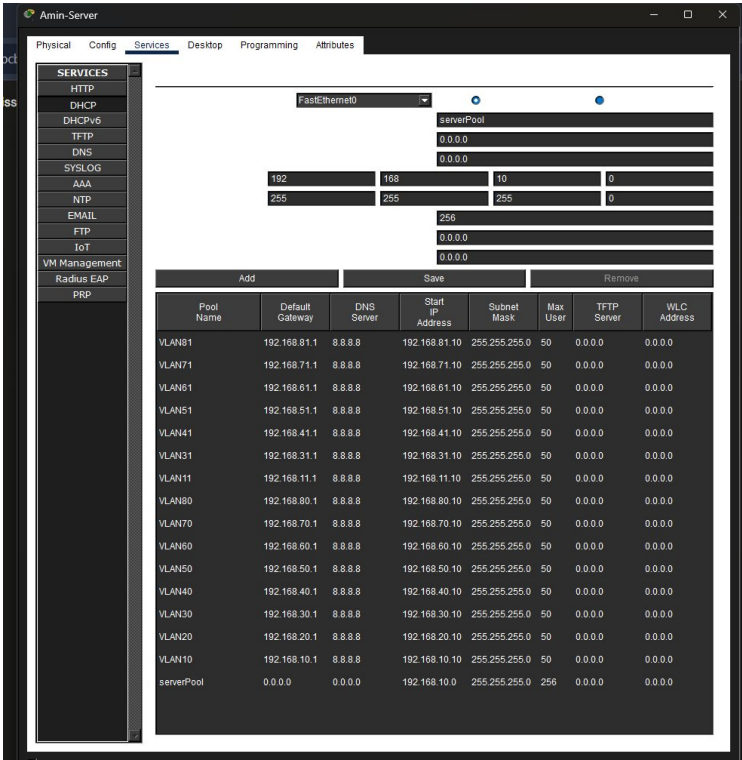
7. DHCP Server Configuration

Amin-Server (192.168.80.10) is configured as the centralized DHCP server for all VLANs across both campuses. Separate DHCP pools are defined for each VLAN with the correct Default Gateway and DNS Server (8.8.8.8).

7.1 Amin-Server — Static IP Configuration

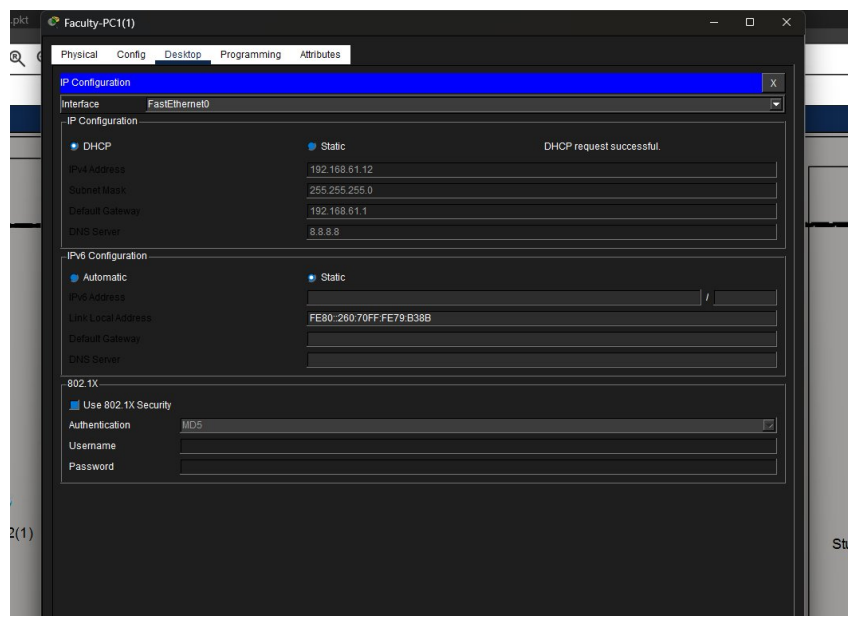


7.2 DHCP Pools — All VLANs



7.3 DHCP Success — Saleem Campus

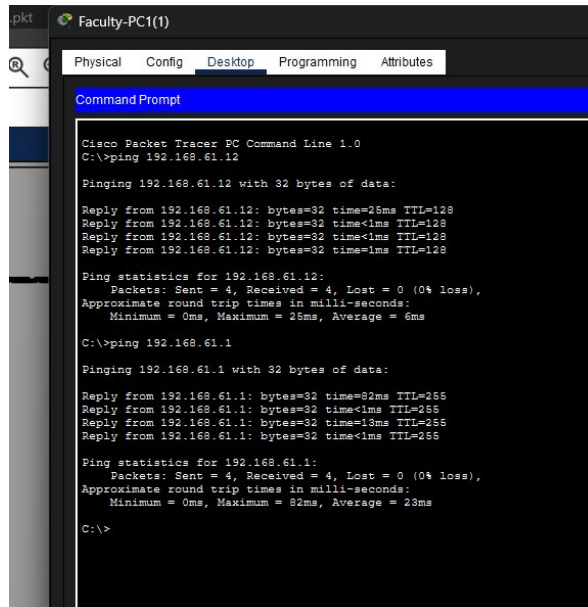
Faculty-PC1(1) on Saleem Campus successfully obtained IP address 192.168.61.12 with Default Gateway 192.168.61.1 from Amin-Server via DHCP relay.



8. Testing and Verification

8.1 Same-VLAN Connectivity Test

Faculty-PC1(1) (192.168.61.12) successfully pinged Faculty-PC2(1) and the default gateway (192.168.61.1) with 4/4 replies and 0% packet loss, confirming intra-VLAN connectivity and correct DHCP gateway assignment on Saleem Campus.



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.61.12

Pinging 192.168.61.12 with 32 bytes of data:

Reply from 192.168.61.12: bytes=32 time=28ms TTL=128
Reply from 192.168.61.12: bytes=32 time<1ms TTL=128
Reply from 192.168.61.12: bytes=32 time<1ms TTL=128
Reply from 192.168.61.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.61.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 28ms, Average = 6ms
C:\>ping 192.168.61.1

Pinging 192.168.61.1 with 32 bytes of data:

Reply from 192.168.61.1: bytes=32 time=82ms TTL=255
Reply from 192.168.61.1: bytes=32 time<1ms TTL=255
Reply from 192.168.61.1: bytes=32 time=13ms TTL=255
Reply from 192.168.61.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.61.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 82ms, Average = 23ms
C:\>
```

8.2 Test Summary

Test Case	Source	Destination	Result
Same-VLAN Ping	Faculty-PC1(1)	Faculty-PC2(1) 192.168.61.12	PASS — 4/4
Gateway Ping	Faculty-PC1(1)	Gateway 192.168.61.1	PASS — 4/4
OSPF Adjacency	Amin-Router	Saleem-Router	FULL/BDR
WAN Connectivity	10.10.10.1	10.10.10.10	Connected
DHCP Assignment	Saleem Campus PCs	Amin-Server	Successful
Sub-Interfaces	Amin-Router	All VLANs	UP/UP

9. Conclusion

This project successfully demonstrates the design and full configuration of a multi-campus university network for The University of Faisalabad using Cisco Packet Tracer. The network implements industry-standard technologies including VLAN segmentation, 802.1Q trunking, inter-VLAN routing via router-on-a-stick, centralized DHCP with relay, and OSPF dynamic routing.

Both Amin Campus and Saleem Campus are fully operational with seven VLANs each covering all university departments. OSPF adjacency is confirmed in FULL state between all routers. DHCP is centrally managed from Amin-Server and successfully distributing IP addresses across both campuses. Connectivity tests confirm that intra-VLAN communication and gateway reachability are functioning correctly.

The design provides scalability for future expansion, security through VLAN isolation, and simplified management through a centralized server infrastructure.